

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10 – STATISTICS 1: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving Question: How can we turn raw data from our school and community into statistics and graphs that help Mama Wanjiku, and other Kenyan decision-makers, solve real problems? In this task you will write a short statistical explanation about the Canteen Queue Puzzle at Mwangaza Secondary School. Answer every prompt in the space under it. Use full sentences, show every step of any calculation, and use the correct statistical words (raw data, tally, frequency, frequency distribution table, mean, mode, median, bar graph, pie chart, line graph). Always link your answer back to Mama Wanjiku's canteen. Remember: Mama Wanjiku sometimes runs out of ugali with sukuma wiki and sometimes has a lot of githeri left over. Across the class's canteen records for the 120 learners, the mean number of ugali portions sold per day was 58, the mode was 60, the median was 57, and ugali made up 47% of all orders.

Part 1: Why did memory fail?

Mama Wanjiku has no written records. She guesses how much to cook from memory. (a) Explain why raw, unorganised data of about 120 meal orders is hard to use, and why her memory-based guesses went wrong. (b) Describe how a tally sheet and a frequency distribution table fix this problem. In your answer, say what a frequency is and what check tells you that your table is accurate.

Write your answer here:

Part 2: Calculating and comparing mean, mode and median

Githeri is the least popular meal. The number of githeri portions sold on five school days was: 8, 10, 7, 10, 10. (a) Calculate the mean, find the mode and find the median. Show your working, including how you ordered the data. (b) The mode and the mean are not the same. Explain why. (c) Which one of the three measures would you advise Mama Wanjiku to use when deciding how much githeri to cook on a normal day? Give a reason.

Write your answer here:

Part 3: Why the median is useful

In one week the canteen sold these numbers of ugali portions from Monday to Friday: 54, 60, 57, 60, 59. On a sports day, the canteen then sold an unusually high 92 portions of ugali, far more than on normal days. (a) Calculate the mean and the median of the five values, then calculate them again when the 92 is added as a sixth value. Compare how much each one changed. (b) Explain how this helps you understand why Mama Wanjiku's memory-based guesses were unreliable. (c) Using the class summary values (mean 58, mode 60, median 57), say what each one tells Mama Wanjiku.

Write your answer here:

Part 4: Choosing the right graph

Mama Wanjiku cannot read a table of numbers during the busy lunch rush. She needs a picture. (a) Name one question about her canteen that a bar graph answers best, one that a pie chart answers best, and one that a line graph answers best. (b) Ugali with sukuma wiki made up 47% of the 120 orders in the class data. Calculate the size of the sector angle for ugali in a pie chart, showing the formula. (c) Explain why the choice of graph depends on the question being asked.

Write your answer here:

Part 5: Report and transfer to the community

(a) Write a short recommendation to Mama Wanjiku (3 to 4 sentences) about how many portions of ugali with sukuma wiki to prepare on a typical day. Use at least two of the statistics from the class data and say which graph you would show her and why. (b) Describe the order of the statistical process we used, from raw data to a decision, and say why the order matters. (c) Choose ONE of these decision-makers: a maize farmer in the Rift Valley, a matatu operator in Nairobi, or a public health officer tracking malaria cases near Lake Victoria. Explain how the same process could help them solve a real problem. This is your answer to the Driving Question.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Organising raw data (tally sheets and frequency distribution tables)	Clearly explains why raw data and memory fail, and accurately describes tally marks, frequency, and the check that frequencies add up to the total, linked to Mama Wanjiku's canteen.	Explains the purpose of a tally sheet and frequency table correctly but misses the accuracy check or a clear link to why memory failed.	Describes tables or tallies vaguely, or does not link them to Mama Wanjiku's problem.
Calculating mean, mode and median	All measures are correct for the githeri data (mean 9, mode 10, median 10) and the ugali data (mean 58 and median 59, then about 63.7 and 59.5 with 92), with data ordered, formulas shown and clear reasons why the measures differ.	Most measures are correct with some working shown; explanation of differences is partly correct.	Several errors or no working shown; little understanding of what each measure means.
Interpreting statistics and graphs	Explains how outliers affect mean and median using the numbers, calculates the pie chart angle correctly (about 169°), and matches each graph type to the question it answers best.	Correct on most graph choices and the angle formula, but explanation of outliers or of the link between the question and the graph is incomplete.	Mixes up graph types or the sector angle formula, and does not explain the effect of extreme values.
Justified recommendation and use of evidence	Gives a clear recommendation of about 57 to 60 portions of ugali using at least two statistics, names a suitable graph	Gives a sensible recommendation using one or two statistics but with limited justification.	Recommendation is unclear, has no statistics, or does not match the data.

	with a reason, and links to the 47% share and githeri waste.		
Transfer to Kenyan community decision-makers	Describes the full process in the right order and applies it convincingly to a Rift Valley farmer, Nairobi matatu operator or Lake Victoria health officer, answering the driving question.	Gives the process in the right order and a reasonable but general community example.	Lists some tools with no order, or gives an example that is unrelated or unexplained.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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